

Perception from Action

Our sensory systems are organized to give us detailed and accurate view of reality, exactly as we would expect if truth about the outside world helps us navigate it more effectively.

—ROBERT TRIVERS¹

The hardest thing is to do something which is close to nothing.

—MARINA ABRAMOVIC²

The eye obeys exactly the action of the mind. In enumeration the names of persons or of countries, as France, Germany, Spain, Turkey, the eyes wink at each new name.

—RALPH WALDO EMERSON (1899)³

1. Robert Trivers, one of the most prominent evolutionary biologists and sociobiologists, made this statement as recently as 2011 (Trivers, 2011). Many other prominent scientists share the view that our perception is veridical and is a window on truth, “veridicality is an essential characteristic of perception and cognition” (Pizlo et al., 2014). David Marr (1982) also expressed the view that humans “definitely do compute explicit properties of the real surfaces out there” but he denied this ability in non-human species. Other theorists make such dichotomy even among senses; for example, that vision is veridical but taste is not (Pizlo et al., 2014).

2. Abramovic (2016).

3. Chapter 5 on Behavior, in Emerson’s essay collection, *The Conduct of Life* (1899), is full of outstanding observations on how human beings and their thoughts are exteriorized by eye movements. “The eyes of man converse as much as the tongues, with the advantage that their ocular dialect needs no dictionary, but is understood all the world over” (p. 173).

In the outside-in framework, the separation of perception from action is an important distinction. Perception-action separation is both intuitive and reasonable in the context of early anatomical studies of the nervous system. In the nineteenth century, François Magendie discovered that the motor neuron fibers that innervate skeletal muscles exit from the anterior root of the spinal cord, while the sensory neuron fibers that carry information about touch from the skin and the contraction states of the muscles from their own stretch-detecting receptors (i.e., proprioceptors) enter through the dorsal root.⁴ That is, distinct sets of neurons deal with body sensation and muscle control. A few decades later, the Ukrainian neuroanatomist Vladimir A. Betz noted that the brain maintains the anterior-posterior segregation of the spinal cord: structures dedicated to vision, hearing, taste, and somatosensation reside largely in the posterior half of the human brain, whereas areas behind the eye, called *frontal structures*, are mainly devoted to motor functions,⁵ in keeping with the anterior motor and posterior sensory organization of the spinal cord. However, while anatomy supports the separation of sensation and motor commands, it has no bearing on the temporal ordering of events, the connections between “sensory” and “motor” areas in the brain, or the postulated causal relationship between perception and action.

Furthermore, while sensation and perception are obviously related, they are different. Some authors use the terms interchangeably; others refer to sensation in lower animals and perception in higher animals. A more rigorous distinction is that sensation is the instantaneous feeling that receptors are being stimulated, whereas perception compares sensation with memories of similar experience to identify the evoking stimulus. Similarly, although “action” and “motor output” belong to the same general category, “action” is a more general term than “motor.” Outputs from the brain can not only affect skeletal muscles

4. Jørgensen (2003); Bell (1811); Magendie (1822).

5. Betz noticed that the anterior (motor) areas in humans contain large pyramidal cells in deep layers, whereas in the posterior part the cells were smaller. This observation led to his best-known contribution, deducing the function of the giant pyramidal neurons of the primary motor cortex (known today as “Betz cells”), which are the origin of the corticospinal tract. “The sulcus of Rolando divides the cerebral surface into two parts; an anterior in which the large pyramidal nerve cells predominate. . . . Undoubtedly these cells have all the attributes of so-called ‘motor cells’ and definitely continue as cerebral nerve fibres” (Betz, 1874, pp. 578–580). In addition to the human precentral gyrus, Betz found these cells in the same location in dogs, chimpanzees, baboons, and other primates, and, based on the “brilliant physiological results” of Fritsch and Hitzig, he concluded that “these cells have all the attributes of the so-called ‘motor cells’ and very definitely continue as ‘cerebral nerve fibres’ (for further details, see Kushchayev et al., 2012). The anterior cingulate cortex of hominids also contains such large neurons, and much speculation has been devoted to the significance of these ‘spindle cells,’ also known as *von Economo neurons*, in intelligence (Allman et al., 2001).

but also exert autonomic effects (e.g., change heart rate) and control endocrine function (e.g., milk production). Furthermore, thought and imagination can also be conceived as actions (Chapters 5 and 9).⁶ Thus, the interaction of the perceptual and action systems is a complex problem.

Most details of the world around us pass without our notice. Stimuli become salient or meaningful through learning and become percepts. But then if perceptual awareness is understood as an interpretation of sensation by the brain, it is an active process. Physiologically, awareness requires that the signal's trace be distributed in many brain structures and linger in neuronal networks for some time.⁷ This fact is interesting because initiating movement also has a volitional component; we are aware of our voluntary actions, as opposed to reflex movements and automatic, well-learned actions, such as walking. Therefore, there is terrain between sensory inputs and motor outputs, a hypothetical central (or "top level") processor. This poorly understood but often speculated about ground is usually referred to by the name of *homunculus*, "decision-maker" or volition.⁸

In the outside-in framework, sensory information is distributed to higher order cortical areas and, consecutively, funneled to the motor areas. The alleged brain route from sensation to motor response is often referred to as the "perception–action loop." Decisions are made somewhere in the unexplained territory between sensory inputs and motor outputs. There is no need for anatomical connections going in the opposite direction.

6. That thought is an action (see also Llinás, 2002) is, of course, a very different idea from the outside-in view that our thoughts are the result of the synthesis of sensory inputs.

7. Many experiments suggest that conscious recognition requires the recruitment of a large number of neurons in an extended, distributed complex brain circuit. According to Libet (2005), this is longer than 0.5 second, which he calls "mind time," supporting a sensory input–conscious decision–action arc model. See criticism of this requirement in Goodale et al. (1986).

8. The little man (or rather a "little rat" in most experiments) is of course only a metaphor for an assumed brain mechanism or, alternatively, the soul. The origin of this homunculus thinking is religion and folk psychology. The soul is often interpreted as an alternative version of the self. Sometime, such as during dreams, the soul is believed to leave the body and even empower it with supernatural abilities. After death, soul and body divorce, and the soul leaves behind the helpless dead body for good. Because the soul has superb interpreting powers, if anything similar exists in the brain, it should have similar interpretive powers. The yin-yang dichotomy of Confucianism reflects a similar yet very different general idea: the yin derives from heaven, whereas yang is from the Earth. Everything is a product of yin and yang. They cannot be divorced from each other. Many, like, Daniel Dennett, liken the homunculus literally to a person watching a movie projected from the retina and other modalities, in what he calls a "Cartesian Theater" where it all comes together; this is in reference to Descartes's dualist view of the mind (Dennett, 1991). The homunculus is often used as a fallacy argument for the infinite regress of logic.

I have already suggested that the brain's primary goal is to generate actions and evaluate their consequences with the help of its sensors.⁹ Combining the historical perspective of cognitive neuroscience (Chapter 1) and the experiments presented later, I will conclude that the correct order is "action–perception" cycle. First, let's examine experiments that lead to my conclusion.

PHI PHENOMENA AND PASSIVE OBSERVERS

Occasionally, our eyes (actually our brains) play tricks on us. Among the best-known illusions is apparent motion or the *phi phenomenon*. In its simplest version, two small balls at two discrete locations on the screen alternate over a short interval (say 60 ms). All humans, and most likely other mammals, see a single ball moving back and forth between the two positions, even though there is no movement whatsoever on the screen.¹⁰ If the two balls have different colors in addition to motion, we also experience an abrupt color change in the middle of the ball's illusory passage between the two locations. This is very interesting because the color change is felt 25 ms or so *before* the second color is actually flashed.¹¹ How can the perceptual effect occur before its physical cause? Phi phenomena have kept philosophers and psychologists busy over the past century offering numerous explanations. They disagree on details, such as whether the mind generates the movement by filling in the missing pieces or the intervening apparent motion is produced retrospectively and somehow

9. Hoffman's interface model of perception (Hoffman, 1998; Hoffman et al., 2005) also emphasizes that we do not simply passively view the world, but also act on it. We can interact with the environment not because we perceive its objective reality but because the world has sufficient regularity that allows for the prediction of our actions on it. However, Hoffman states that "having a perceptual experience does not require motor movements" (Hoffman et al., 2005). I agree that this may be the case in an already movement-calibrated brain, yet I maintain that without action calibration at some stage of life perception does not arise.

10. The phi illusion was discovered by the German psychologist Max Wertheimer (1912). For an extended discussion, see Kolers and von Grünau (1976) or Dennett and Kinsbourne (1992). Many contemporary neuroscience researchers use illusions and subjective feedback about them as tools to probe neuronal responses (Dehaene and Changeux, 2011; Koch, 2004). The title of Giulio Tononi's excellent and entertaining book on consciousness-related topics is *Phi* (2012).

11. There are many interesting practical and theoretical implications of the phi phenomena. In a movie theater, we do not perceive a rapid flipping of still frames at 24 frames per second. Instead, we have the illusion of continuous movement without interruptions. Neon light advertisements also use the same principle to create motion perception.

projected backward in time, but nearly all of the theories call upon consciousness as *explanans*, the thing that explains, rather than the phenomenon to be explained.

Phi phenomena, as well as ambiguous pictures, Rorschach patches, and other illusions, are favorite territory of consciousness researchers. This research program often focuses on perception, asking, for example, how the brain can extract and bind sensory features to produce a unified object. But this program stops short of explaining the perception–action cycle¹² because it says nothing about action. This neglect creates a huge gap between perception and action that requires filling it with mental constructs such as volition and decision-making. This separation largely explains why research on perception and action have taken independent paths in neuroscience.

If action is a prerequisite of perception, as I claim, then there should be no perception without action. You can show that this is indeed the case by fixing your gaze steadily on a stationary target. No matter what surrounds the gaze target, even if pink elephants are running around, it will all disappear after just a few seconds. This never-failing demonstration is known as the *Troxler effect*.¹³

Another way to probe the primacy of action is to get rid of its consequences altogether. One such manipulation is called *retinal stabilization*. In an early, ingenious instantiation, a suction cup with a mirror on it was attached to the cornea. (Today, we would call it a contact lens.) When a picture or movie is projected onto the mirror, you can no longer investigate the scene with your eye movements because the image on the retina moves with your eyes, in the same direction and at the same speed and amplitude. As you can guess by now, without the brain's ability to obtain a “second opinion” by eye movement investigation, vision ceases.¹⁴

12. For a discussion of the involvement of higher order cortical areas, see Quintana and Fuster (1999).

13. Named after Ignaz Troxler, a Swiss physician who first demonstrated the effect (Troxler et al., 1804). We must fix our gaze to inspect the details of a scene, but if we fixate a while, it fades from our brain. See it yourself: https://en.wikipedia.org/wiki/Troxler%27s_fading. One explanation of the Troxler effect and related illusions is neuronal “adaptation” in the retina or further up in the visual system, without the actual identification of the cellular and circuit mechanisms of the adaptation (Martinez-Conde et al., 2004). The moment you move your eyes, sight returns. Zen Buddhism effectively exploits this feature of the visual system for meditation. In the famous zen Rock Garden (Karesansui) in the Ryoanji Temple in Kyoto, the meditator can focus on one of the fifteen rocks arising from flat white sand and make the rest of the (visual) world disappear. It works well; I tried it.

14. The most popular citation for the stabilization experiments is Yarbus's book (1967). Alfred Yarbus was a Soviet psychologist who recorded eye movements in humans while they viewed objects and scenes. He showed that the eye movements reliably track the observer's interest in

ACTION SPEED LIMITS PERCEPTION SPEED

Instead of asking why the brain can be fooled by artificial or rare patterns, perhaps it is more productive to ask why the brain reliably produces illusions.¹⁵ As autoshaped behaviors are not mistakes (Chapter 1), illusions are not mistakes either. They are due to the brain's ability to efficiently extract important features from garbage. Some distinctions are of paramount importance, whereas we fail to observe most of the enormous variability in the environment.¹⁶ Perception is not the veridical representation of the objective world even in case of vision, to which sense the human brain devoted a very large part of its real estate. Even the most sophisticated observer, without the knowledge that the color balls in the phi illusion do not move, cannot tell whether the brain is faithfully following the motion of a ball or whether two stationary balls are being flashed in a rapid sequence. There is no way to distinguish the two possibilities because brain mechanisms to track such fast changes consciously do not exist.

So why not have a faster perceptual system? If the primary goal of the brain were to perceive and process, it would be odd for evolution to compromise speed. The answer may lie in the mechanics of our body. Conditionally accepting my conjecture that the brain's fundamental goal is to produce action, there is no reason to invest in speed if the muscle system that it controls is slow. The contraction speed of vertebrate muscles is determined by the properties of *myosin*, a contractile protein that is largely preserved across all mammals independent of body size. Therefore, small and large brains must deal with problems at the same temporal scale. In the phi illusion, jerky or saccadic movements of the eye are needed to refocus the color vision region (fovea) of the retina from

the details of the objects. In the case of the human face, the investigation of eyes, mouth, cheek, or nose regions depends on what emotions are being identified. See also Riggs and Ratliff (1952) and Ditchburn and Ginsborg (1952). There is a resurgence of interest in eye-tracking techniques today, perhaps due to the recognition that they provide useful information about the "mind's opinion and intentions." Wearable systems can identify one's interest in particular details of scenes and advertisements with high precision. Every major technology-oriented company, from Sony to Google, offers such devices.

15. Visual areas V1 and MT may be involved in phi phenomena as they respond to real movement and apparent movement (e.g., induced by alternating flashing of stationary stimuli). Neuronal spiking in these cortical regions varies systematically with spatial and temporal properties of the stimuli that closely parallel the psychophysically defined limits of apparent motion in human participants (Newsome et al., 1986).

16. Visual artists exploit illusory perceptual experiences for aesthetic goals. Victor Vasarely, who was born in my hometown of Pécs, Hungary, and other painters create impossible three-dimensional objects on planar surfaces. The impossibility becomes apparent only after carefully scanning the continuity of the lines. Or think of another painter: Maurits Escher.

one ball to the other. These ballistic eye movements occur three to four times a second on average and are the fastest movements in the human body (100–300 degrees/s or 20–30 ms during reading).¹⁷ The skeletal muscles are considerably slower than those that move the eyeballs. Nobody can move his fingers faster than 20 taps per second, not even Franz Liszt.¹⁸ Even if an imagined super brain could perceive, decide, and send a movement command in less than a millisecond, it would have little benefit for the muscles, which must temporally coordinate their actions over a sluggish tens to hundreds of milliseconds. From this evolutionary perspective, we have one more reason to believe that perception is secondary to action.

Perception–Action or Action–Perception?

The clever skeptic, of course, can argue that there is no reason to prioritize the “chicken and egg” two-way problem of action–perception because it is a cycle anyway. An often-used argument in favor of the perception–action cause–effect direction is the existence of simple reflexes, such as the *patellar reflex*. When a neurologist strikes the patellar tendon with a reflex hammer, it stretches the muscle spindle receptors in the quadriceps muscle. The muscle receptors are peripheral axon processes of the neurons in the dorsal root ganglion of the spinal cord, and they convey the stretch-induced change to the ventral horn neurons of the cord, which in turn send their axons to innervate the same quadriceps muscle. Every time the neurologist strikes the tendon, the monosynaptic response is activated and a kick is observed. At the same time, another axon branch of the neuron in the dorsal root ganglion sends the same spike message to another neuron in the spinal cord which, in turn, conveys the information to the thalamus. In line with the Bell-Magendie rule, so the argument goes, the sensorimotor reflex must be the evolutionary precursor of perception. But who is doing the sensing? The neuron in the ganglion, the ventral horn motor

17. Myosin is a contractile protein found in skeletal muscles. The contraction of human myosin is only twofold slower than the contraction of myosin in the rat, which is more than 100-fold smaller (Szent-Györgyi, 1951). For eye-movement speed in humans, see Fischer and Ramsperger (1984). Muscles are the second cell type in the body that can rapidly change their membrane potentials (and generate action potentials). Several key molecules in muscles have similar functions in neurons. Both muscles and neurons can generate energy by both aerobic and anaerobic metabolism (e.g., Buzsáki et al., 2007).

18. In skeletal muscles with sensory feedback, fast afferent processing is performed by two large side appendages of the thalamocortical systems: the olivocerebellar system and the basal ganglia. In his best-selling book (*I of the Vortex*; 2002), Rodolfo Llinás eloquently describes the evolutionary reasons that the brain should lag behind the body's speed.

neuron, or the thalamus? What if some disease destroys the motor neurons but leaves the sensory pathways intact from the muscles and skin to the spinal cord, thalamus, and the rest of the brain. Of course, no stretch reflex will occur. But the issue is whether the patient will sense the strike of the hammer. I vote for a no answer because the strike-induced neuronal activation has no meaning to the brain since it has not been grounded.

Let's put my argument into a broader evolutionary perspective: an animal that can only act has a higher probability of survival than an animal that can only perceive. Sensing, or being aware of sensing, has no utility whatsoever unless the organism can act. On the other hand, movement can be useful even without any sensory information. In seawater, where our ancient animal ancestors lived, with abundant food around, rhythmic movement is sufficient to feed a simple creature. Once a movement control mechanism is in place—and only after this step—it makes sense to develop sensors that more efficiently guide movements toward food or shelter to improve the chances of survival. This argument is supported by another observation that the nervous system and muscles emerged together in the jellyfish.¹⁹ Only through doing something do animals have some purpose of being.

From the perspective of evolution, therefore, there is a fundamental distinction between “perception–action” and “action–perception.” If perception were the primary objective of brain design, many of our perceptual systems would appear to be built poorly. Movement, on the other hand, has to have useful consequences to call it an “action.” Disconnecting its consequences on sensing the surrounding world breaks the action–perception loop and renders the system useless.

Such interruption of the cycle can occur naturally. During rapid eye movement (REM) sleep, we temporally lose our body, as our muscles cease to be controlled by the nervous system. When we wake, we immediately get our body back. Occasionally, these brain functions get de-coordinated in time, and we wake up a few seconds before muscle control is reinstated. This terrifying feeling is known as *sleep paralysis*. The brain is confused because stimuli impinge on our sensors, but they make no sense and induce no meaning since we cannot act on them. Demonic voices are heard, grotesque incubi dance on our chest, and we feel threatened. We are not dreaming but fully alert. These hallucinations, like the illusions we discussed earlier, are an inevitable

19. Llinás (2002) points out an even more striking example of the primacy of action in the tunicate, a marine invertebrate animal. In their larval stage, tunicates resemble a tadpole and move around happily looking for a home base. As soon as they find a suitable place to settle, they metamorphose into a barrel-like sedentary adult form. In this sessile existence, there is no need for movement, so tunicates digest most of their brains.

consequence of brain-body-environment interactions.²⁰ We connect to the world not through our sensors (although they are essential) but through our actions. This is the only way that sensation/perception can become “grounded” to the real world as experience (Chapter 1). The distance between two trees and two mountain peaks may appear identical on the retina. It is only through walking and moving one’s eyes that such distinctions can be learned by the brain. Again, I reiterate that the brain’s main function is not veridical perception and representation the objective world with its mostly meaningless details, but to learn from the consequences of the brain’s actions about those aspects of the environment that matter for particular goals, such as reduction of hunger.

Recent anatomical and physiological experiments provide abundant support in favor of the critical role of action in perceptual computation. Primary sensory areas, and especially areas designated as “higher order sensory,” receive numerous inputs from brain regions defined as “motor.” In fact, beyond cortical areas which receive direct, thalamus-relayed inputs from the senses and which send axons to the spinal cord, all other areas show very strong and widespread interconnections.²¹ Neurophysiological experiments in rodents document that more than half of the variability of neuronal activity across the entire cortex, including those we refer to as primary sensory, can be explained by self-generated action parameters such as body motion, whisker movements, orofacial movements, and pupil diameter changes. Such strong action-modulation of brain activity belittles responses to sensory stimuli. The involvement of wide brain areas in action suggests that nearly every cortical neuron receives two types of afferents, conveying both action and sensory inputs, although in different proportions.²²

20. I was scared to death during my first sleep-paralysis experience. But safe recoveries from the momentary early morning attacks allowed me to internalize the experience (as discussed in Chapter 5), and they no longer bother me. In other words, I “grounded” these experiences by assigning a neutral meaning to them. You may have had similar adventures; nightmares occur in about one-third of the adult population (Ohayon et al., 1999) though usually during non-REM sleep, which is different from sleep paralysis. Horrific experiences in rare cases of anesthesia arise from the same mechanism: the brain can (mis)interpret sensory inputs without its ability to act. Anxiety and panic attacks may involve similar mechanisms due to our inability or perceived inability to change a situation and to be in charge.

21. Petreanu et al. (2012); Economo et al. (2016); Chen et al. (2018); Han et al. (2018); Harris et al. (2018).

22. Stringer et al. (2018) concluded that “an understanding of the function of sensory cortex will likely be impossible without measuring and understanding its relation to ongoing behavior, beyond easily-characterized measures such as running and pupil dilation.” See also Musall et al. (2018).

SHORT-CIRCUITING THE ACTION–PERCEPTION CYCLE

When we shout loudly near someone's ears, she may have difficulty hearing for a few seconds. In contrast, we can resume normal conversation right after our own shouting is heard by our ears. We are protected in that case by multiple mechanisms in our auditory system, as the Greeks speculated long ago. Today, we have a name for it: *corollary discharge*. As the name implies, this ancillary activity occurs simultaneously with the action output. The action-initiating circuits of the brain send action potentials not only to the downstream motor pathways but also synchronously to other areas in the brain. This secondary activity provides a feedback-reporting mechanism for self-organized action. "I am the agent that brought about change in the sensors."

A particular sensory modality may be activated when a stimulus unexpectedly occurs or when the animal is actively searching for the stimulus or expecting it. These two types of stimuli evoke different responses, like the feelings when you touch or are touched. An unexpected stimulus can induce either a generalized response, such as a startle reaction in response to a loud noise, or an orienting response.²³ The orienting response is the brain's active attempt to learn more about an unexpected event.

As discussed in Chapter 1, neurons cannot interpret the relevance of the signals conveyed by sensory inputs because they cannot ground their response without some independent verification. For neuronal networks to interpret the world, they need two types of information for comparison. The extra information can be supplied by the movement-induced changes of the brain's sensors. Only by comparing two signals, one of which is grounded by movement or previous knowledge (which was also generated by action at some point in life), can the brain figure out what happened out there. This distinction is easy to demonstrate by closing one eye and moving the open eyeball from the side

23. The term "orienting reflex" was coined by Ivan Pavlov and studied most extensively by Evgeny Sokolov (1960, 1963). Sokolov noted that if a novel signal is not followed by reward, the animal quickly learns to ignore it, a process he called *habituation*. Interestingly, Sokolov used the same comparator circuit as used in the corollary discharge model (Figure 3.1) to describe how the brain (particularly the hippocampus) can detect whether something is novel or not. This is perhaps not surprising. When an eye saccade occurs, the brain makes a prediction about the expected content of the scene after the saccade based on the current state of its sensory knowledge (a Bayesian prior; Chapter 13). If nothing new happens, no sensory change is registered. However, if a novel thing appears during the saccade, there is a mismatch between expected and actual inputs (i.e., if the prior model is erroneous, the model is updated). The process of updating can be called perception. Behaviorally, a mismatch between the predicted and detected states of the sensors will trigger an action: the orienting response.

with your finger. The entire visual field will appear to move at the pace of your finger's movement. In contrast, when you move your head back and forth or scan the world with your eyes, producing a similar pattern of stimulation on the retina, the world appears stationary even if you blink or make saccadic eye movements. Critically, if your eye muscles were paralyzed for some reason, and you intended to make an eye movement to the right, then you *would* observe a shift of the visual field in the same direction.²⁴ What would be a proactive brain mechanism to compensate for such confusing shifts?

The Proactive Corollary Discharge Mechanism

Distinguishing the real and perceived worlds has entertained both philosophers and artists prior to neuroscience. Perhaps the Italian painter Filippo Brunelleschi was the first to test it experimentally. On a wooden panel about twelve inches square, he painted the Baptistery of Florence in its surroundings. In the center, he made a hole and invited the viewer to look through the back of the painting while he held a mirror. Brunelleschi carefully positioned the painting (i.e., its reflection in the mirror) so that the real baptistery and the painting lined up perfectly. Through the mirror, the real world looked more two-dimensional and the painting more three-dimensional, thus the viewer could not distinguish easily between the two mirror reflections. His goal was to demonstrate that only by moving the head back and forth can reality and image be distinguished, making and unmaking the viewer's own world. It is the viewer's voluntary action of head movement in this artistic peep-show that can distinguish real change from apparent change.²⁵

Erich von Holst and Horst Mittelstaedt, working on fish and invertebrates in Germany, and Roger Sperry, studying the optokinetic reflex in humans at the California Institute of Technology, went further than Brunelleschi. They wondered whether a motor command exits the brain unnoticed or leaves a trace behind.²⁶ They came to the conclusion that there must be brain mechanisms that separate self-generated stimulation from externally induced stimulation

24. This is exactly what patients with paralysis of the eye muscles see. Herr Professor Kornmüller (1931) famously anaesthetized his own eye muscles to investigate what happens when the volitional act is blocked. He reported an apparent shift of the visual field, which could not be distinguished from a similar magnitude of true shift of visual input.

25. As described in Wootton (2015, p. 165).

26. The paper by von Holst and Horst Mittelstaedt (1950) is an extraordinary read. In thirty pages, the entire history of motor control and sensory physiology is exposed, many experiments from flies to fish are presented, and the findings are summarized in quantitative models. The authors alert the reader that the brain of even the simplest creatures consists of more than "simply a set of connecting cables between receptors and muscles!" They not only demonstrated

of the brain's sensors. This hypothetical mechanism assumes that a copy of the motor command signal leaves an image of itself somewhere in the brain. Sperry called it "corollary discharge," while the German duo referred to it as "reafferenz," both of which likened it to the negative of a photograph. In turn, the stimulation from the sensor during the movement is the positive print that, when united with the negative, makes the image disappear. Experiments by Sperry, von Holst, and Mittelstaedt were among the first to demonstrate that action directly influences sensation. Neurons giving rise to an eventual motor command send a copy to neurons dealing with the sensor-supplied signals. These latter neurons (call them *comparators*) can compare what comes in and what goes out from the brain²⁷ (Figure 3.1).

The corollary discharge (i.e., the mechanisms that allows the sensory system to be informed about an action command) is an invention of the brain. No such internal reafferentation exists from the ventral horn motoneurons to the sensory side of the spinal cord. So Betz's intuition is only partially right. There is indeed a geometric separation of the posterior "sensory" and anterior "motor" areas in the brain. However, their way of communication is very different from that of the spinal cord. The spinal cord is a modular system, with each module utilizing pretty much the same algorithm and intersegmental coordination. It is mostly about reflexes and locomotion supported by a large number of neurons.

that an insect can discriminate between its own movements and movements in its environment but were confident enough to generalize their findings: "the reafference principle applies throughout the CNS [central nervous system] from the lowest phenomena (internal and external control of the limbs, relations of different parts of the body to each other) to the highest (orientation in space, perception, illusions)." Compare this with Sperry's description "[The] kinetic component may arise centrally as part of the excitation pattern of the overt movement. Thus, any excitation pattern that normally results in a movement that will cause a displacement of the visual image in the retina may have a corollary discharge into the visual centers to compensate for the retinal displacement. This implies an anticipatory adjustment in the visual centers specific for each movement with regard to its direction and speed . . . with the retinal field rotated 180 degrees, any such anticipatory adjustment would be in diametric disharmony with the retinal input, and would therefore cause accentuation rather than cancellation of the illusory movement" (Sperry, 1950, p. 488). The essential features of corollary discharge are also the foundation of predictive coding (Rao and Ballard, 1999; Kilner et al., 2007) and motor control theory (Shumway-Cook and Woollacott, 1995; Wolpert et al., 1995; Kawato, 1999; Grush, 2004).

27. The corollary discharge mechanism represents a basic pattern of loop organization of the brain. Output patterns inform input analyzer circuits. We will discuss extensions of this fundamental mechanism in multiple chapters. The comparator circuit has been long viewed as a fundamental mechanism of many brain functions (Sokolov, 1960, 1963). MacKay's (1956) epistemological automaton compares actual and expected sensory inputs. Excitatory recurrent feedback circuits might be viewed as an internalized version of the corollary discharge mechanism.

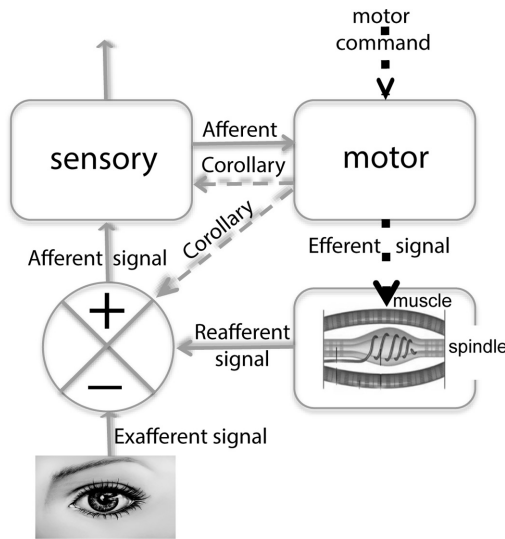


Figure 3.1. Schematics of the corollary discharge mechanism. Motor command signal is sent from the motor areas to eye muscles (efferent signal). At the same time, corollary discharge (*dashed arrows*) is also sent to comparator mechanisms in the sensory system. The comparator performs a subtraction or divisive normalization on the external (exafferent) signal determined by the corollary discharge. In addition, the magnitude of the reafferent signal from the tension sensor in the muscle can exert a delayed affect on the sensory signal. Projections from motor cortex to sensory cortical areas are a common architectural feature in all mammals.

In fact, the volume of the spinal cord in the elephant is larger than the brain of a macaque monkey. So the “smartness” of the brain is due not only to large numbers of neurons and synaptic connections. What also matters is how those connections are organized. The corollary recurrent loop from the output to the input is such a key motif of the cortex. I will return to recurrency, positive feedback, and corollary discharge in several chapters to argue that these mechanisms are the most important grounding processes of cognitive operations.

Two Types of Corollary Discharge Mechanisms

There are at least two ways to distinguish the sensory stimulation produced by one’s own movement from real motion in the surrounding world. The first one can be called *adaptive filtering*, an example of which can be found in crickets. Have you ever tried to catch a chirping cricket? Crickets are wildly loud, but when you move toward them, they detect your motion and become quiet. For this reason, farmers in rural Japan used crickets as sentinels to protect their

households. They put them in small cages where they sang during the dark hours of the night, except when they sensed something moving. The ensuing silence then served to alert the householder.²⁸ Crickets do not have ears, but they are equipped with a pair of tympanal organs on their legs. This super-sensitive device can detect vibrating air molecules (sound), which provides their primary alarm mechanism to avoid predators. In addition to listening, male crickets also “sing”; that is, they have a file and scraper body part on their wings that makes a loud *chirping* noise. Their 100-decibel chirps are several-fold louder than the sound of your footsteps, yet the cricket can be fully alert while chirping. Its brain counteracts the massive influx of auditory, proprioceptive, and mechanoreceptive information during sound production by inhibiting its central auditory neurons with a corollary discharge synchronously with sound production. Remarkably, in the cricket’s brain, the mechanisms responsible for this forward signaling are accomplished by a single inhibitory neuron. When this corollary discharge neuron is active, it selectively inhibits the response of auditory neurons associated with self-generated chirps. A subtraction or division/normalization process (Chapter 11) in these auditory neurons removes the loud self-generated vibration signal yet maintains the cricket’s ability to detect much quieter external stimuli.²⁹

Distinguishing real changes in the environment from apparent sensory changes due to one’s own movement can be also accomplished through a second mechanism that could be called “time division” because corollary discharge and signal detection occur at different times. For example, when you blink or produce saccadic eye movements, the input from the world to the retina disappears or gets blurred, but we do not notice that. An ideal observer could “see” everything around itself all the time, but in the real world the “seeing” is a compromise between spatial resolution and speed. A fly’s eye and a digital camera both have a certain number of pixels to work with. The camera typically scans the target pixel by pixel, and its effective resolution is determined by the number

28. As told by Dethier (1987). His essay is worth reading for both scientific insight and the aesthetic beauty of the narrative.

29. In a remarkable series of studies, Poulet and Hedwig (2006) used simultaneous intracellular recordings of auditory neurons in the prothoracic ganglion, where primary auditory signals are processed, while searching for the corollary discharge in the mesothoracic ganglion, which houses part of the chirp pattern-generating network. The dendrites and cell body of the inhibitory corollary discharge neuron are surrounded by the chirp pattern generator neurons, which excite it, and its extremely large axon tree forms numerous synapses with the auditory neuropil in the prothoracic ganglion. The neuron generates bursts of spikes in synchrony with each chirp but does not respond to external sound. Its destruction has no impact on chirp production, so its sole job description is to provide corollary discharge to the auditory neurons. To date, this is the clearest mechanistic description of such a mechanism.

of pixels the processor handles in a given amount of time. The compound eye of the fly has approximately 3,000 pixels, and, by flying fast (perhaps fifty times faster than we walk, relative to body size), the fly generates an image shift on the retina, called *optic flow*, that scans a large part of its environment. The fly's erratic-looking zig-zag flights are actually stop-go saccadic patterns that actively shape its visual input by enhancing depth resolution and distinguishing figures from the background.

Other creatures scan the world differently. Frogs lack eye muscles, but respiration moves their eyes rhythmically. The preying mantis moves its entire head in saccadic scans. Spiders, with their body anchored to their net, protrude their eyes and rotate them like miniature periscopes to view the world. Vertebrates with superior brain processing power exploit both spatial resolution and scanning strategies. The eyes of primates, including humans, have many cones in the foveal region for accurate vision and many more rods surrounding the fovea to detect motion. Primate eyes can move either slowly (explorative) or quickly (saccadic). Like spiders, we can stay stationary yet monitor a very large part of our surroundings as long as our brain moves our eyes. During saccadic eye movements, the visual scene does not get blurred. Instead, it is suppressed momentarily by postulated corollary discharge. The visual field appears to be at one place or another, without experiencing the in-between scan. You can check this by looking into a mirror. You see yourself continuously and do not notice your saccadic eye movements. The suppression mechanism alerts visual processing areas in the brain that the forthcoming disruption is the result of your brain's command, not a change in the visual world.

In the primate brain, the neuronal hardware of this forward signaling is a bit more complicated than in the cricket, but the principle is the same. Neurons in the frontal eye field area of the neocortex are good candidates for the job because they receive both visual and corollary discharge information about eye saccades from the medial dorsal nucleus of the thalamus. After comparing the visual input with the grounding information about eye movement, frontal eye field neurons inform their targets whether the stimulus remained stable or moved, the strength and timing of their corollary discharge response, and the magnitude of stimulus translation if the scene moved during the saccade. Inactivation of the action path, for example by infusing local anesthetic in the thalamic medial dorsal nucleus to abolish spiking activity locally, will result in a failure to inform the visual neurons about the necessary grounding information. As a result, frontal eye field neurons supply nonsense information to their targets. Under such conditions, the experimental monkey reports that the visual scene jumps with each saccade just as humans feel after eye muscle paralysis. In further support for the role of the frontal eye field, experiments in monkeys show that microstimulation of this area enhances the responsiveness of visual

cortical neurons with spatially overlapping receptive fields.³⁰ The just-discussed experiments and observations indicate that the self-motion-generating circuits provide a prediction signal for the sensory system so that the expected signal can be compared with the one conveyed to the brain by the sensors.

The Illusion of Continuous Vision

Sampling of the visual world is not continuous because it is interrupted by saccades, with each saccade causing a loss of up to 10% in sampling time. However, this loss has important advantages. First, blurred vision is prevented because during the saccade the visual input is suppressed. Second, corollary discharge is an important timing signal that helps coordinate neuronal activity beyond the visual system. Third, suppressed spiking in several types of visual neurons during the saccade allows them to replenish their resources, for example by restituting dendritic sodium and calcium ion channels inactivated during intense spiking. As a result, after the saccade, the visual system is transiently more sensitive to stimulation, a considerable gain.

Action in the form of muscle activity (again!) may be the evolutionary origin of intermittent sampling of sensory inputs. The muscle's fastest reaction is a twitch followed by a refractory period, during which it cannot be activated. This mechanical constraint may explain why sensors adapted to such a regime. Regardless of its origin, intermittent sampling is advantageous for transmitting information. First, intermittency introduces a way to chunk the information, similar to spaces between words. Thus, every neuronal message can have a beginning and stop code (Chapter 6). Second, intermittency simplifies how information processing can be coordinated across brain areas because it introduces the means of generating a clear time reference frame that can be shared by all mechanisms involved. Blind people discovered the supremacy of intermittent

30. Neurons in the parietal cortex (particularly in the lateral intraparietal area) can also function as corollary discharge comparators and drive neurons in the frontal eye field. Other brain regions, including the lateral geniculate body, superior colliculus, and even early visual areas V1 to V4 appear to be involved. In addition to suppression of retinal inputs, many interesting things happen in the brain around the time of saccadic eye movements, including compression of space and time and displacement of part of the world depending on the direction of the saccade (see Chapter 10). The magno- and parvocellular pathways in the early stages of the visual system are affected differentially. It is mainly the magnocellular pathway of the lateral geniculate body that is suppressed by saccades and informs the motion centers. In addition to corollary mechanisms, visual motion caused by the eye movement itself can contribute to masking vision. Duhamel et al. (1992); Umeno and Goldberg (1997); Sommer and Wurtz (2006); Crapse and Sommer (2012); Moore et al. (2003). For reviews, see Ross et al. (2001); Crapse and Sommer (2008).

sensory sampling by realizing that forms constructed from raised dots were more easily discriminated by the fingertip than continuous raised lines.³¹ Sensing text with fingers also begins with movement.

PERCEPTION AND ACTIVE SENSING

Not everyone has the luxury of saccadic eye movements. A research participant named AI was unable to move her eyes since birth because of fibrosis in her eye muscles. Yet she had no complaints about her vision. She could read and write, although her reading speed was slow. She accomplished this feat, compensating for her lack of eye movements, by making head saccades. These saw-tooth movements to her right had an average duration of 200 ms. At the end of the line, her head jerked back to the left side of the page to read the next line. During picture viewing, AI's head movements scanned the important details of the picture, similarly to the way we deploy eye saccades to extract interesting features from a scene (Figure 3.2). By actively moving her head (and consequently her eyes), AI could enhance her visual sensing ability.³²

Active sensing refers to a brain-initiated search as opposed to the response to an expected event.³³ Perhaps another word, “observation,” equally well captures the same process. In the real world, stimuli are not given to the brain. It has to acquire them. The sensitivity of sensors depends, in part, on the effectors that can move them and maximize their efficacy. It is a bit like the relationship between a camera and its user. The camera takes the photo, but what is captured depends on the action of the user. Unexpected stimuli, when they occur independently of movement, trigger search behavior and optimize the sensors. The physiological foundation of such active sensing is also corollary discharge.

Active sensing occurs in two forms: the entire body can translocate (e.g., the fly's solution for vision), or sensors are moved locally (e.g., the spider's solution

31. Louis Braille, who was blind from childhood, went on to create the Braille tactile writing system, a sort of parallel Morse code.

32. Gilchrist et al. (1997).

33. Ehud Ahissar at the Weizmann Institute refers to active sensing as “closed-loop” sensing, in contrast to the perception–action semi-loop, in which the interaction between the sensory organ and the environment is the first step (Ahissar and Assa, 2016). James J. Gibson's affordance theory can be considered as a forerunner of active sensing. According to his theory, not only is the world perceived in terms of object shapes and spatial relationships, but these things also offer intuitive possibilities for action (affordances). The environment “offers the animal, what it provides or furnishes, either for good or ill” (Gibson, 1979). While Gibson tried to eliminate the decision-maker between perception and action, he still thought that perception drives action.

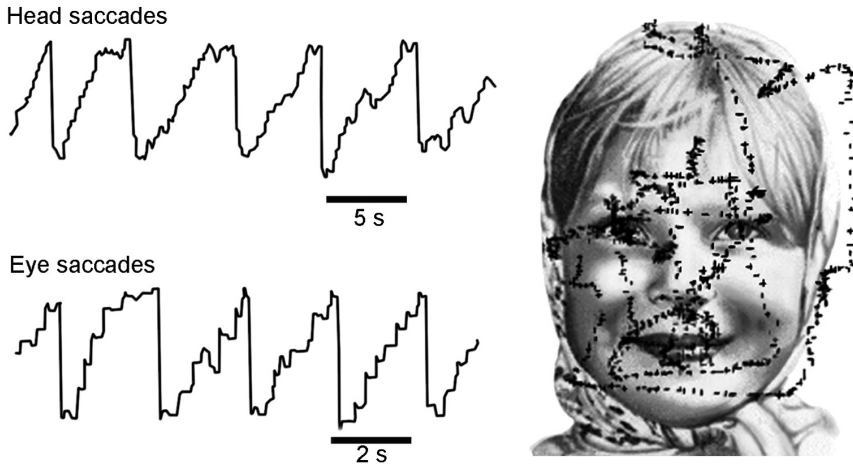


Figure 3.2. Head movements of patient AI with paralyzed eyes (head saccades) and eye movements from a control subject (eye saccades) during text reading recorded using a head-mounted search coil. Although AI is considerably slower overall, and her head stability is not as good as that of the control subject, her saccade strategy is the same as that of the normal subject. Right, AI's head movements while viewing a picture. Redrawn after Gilchrist et al. (1997).

for vision), but the purpose is the same. The two strategies can be exploited by the same animal. For example, when a hunting dog is taking a bird scent sample on the ground, it sniffs the grass rhythmically, but to chase the same scent in the air, it runs fast with its nose up to create a scent flow. In either case, the physiological foundation to make sense of such active sensing behavior is corollary discharge.

Mechanisms of Active Sensing

Researchers understand the neuronal mechanisms underlying certain forms of active sensing. Some animals can both sense and generate electric fields. The weakly electric fish (*Eigenmannia*) actively probes its environment with electricity. It has a specialized structure called the *electric organ* in its tail, which emits electric fields. It also has electroreceptors called *tuberosus receptors* in its skin that sense electric fields for object detection (*electrolocation*) and communication with other electric fish (*electrocommunication*) in muddy waters.

In electrolocation, the fish's brain analyzes the frequency and phase difference of its emitted sinusoid electric field and the version returned from objects. When two fish are near each other, the frequency difference of the fields emitted by each

is determined by the combination of the two fish's discharge patterns. Thus, the problem of sorting out self-generated electric field from fields produced by other fish is the same as that of touching something versus being touched—only the medium is electrical, not mechanical. The solution of the brain is also similar: a circuit analogous to the corollary discharge signal comparator. The firing pattern in the electric organ is controlled by motor neurons that are driven by the motor command nucleus. The axons of the neurons from the command nucleus bifurcate, with one pathway going to the medullary relay nucleus whose axons project along the spinal cord and excite electromotoneurons that generate the electric organ discharges. The other collateral pathway innervates the bulbar area, which contains the corollary discharge comparator circuit (Figure 3.1). This circuit is analogous, albeit different in its details, to the corollary circuit of the cricket.³⁴

Echolocation, also called *bio-sonar*, is another well-known active sense that works by emitting sound or ultrasound calls into the environment and analyzing the returned echoes to locate objects and prey. Bats and marine mammals are the animals best known for their sonar sensing. Whales use a series of clicks during their dives. The precision of bat echolocation is good enough to identify mosquitoes even when both the insect and the bat are flying. The sound localization mechanisms in the brainstem act by detecting the time differences of sounds between the two ears. The self-vocalized sounds are strongly attenuated between the cochlear nerve and the inferior colliculus, early stations of the auditory system, as well as in the auditory cortex.³⁵

The neuronal mechanism used in echolocation is similar to our ability to use self-generated sounds to tell whether we are in a small or large room. Normally, we do not use this option much because we have the luxury of vision. However, this ability can be enhanced if needed. Some blind individuals can replace their eyesight by making loud footsteps, tapping a cane, or producing mouth clicks and listening to the returning echoes. These active sensing methods are good enough for crossing roads, even riding a bicycle or rollerblading. Some exceptional blind individuals can identify an object's distance, size, and texture, such as the difference between a metal fence and a wooden fence.³⁶

34. Walter Heiligenberg's book is a masterful combination of ethology, electrophysiology, and modeling on electric sensing in *Eigenmannia* used for jamming avoidance (Heiligenberg, 1991). For an update, see Chacron (2007).

35. Suga and Schlegel (1972); Hechavarria (2013).

36. Echolocation is a trainable skill that can provide blind people with self-reliance in their daily life. Ben Underwood and Daniel Kish, two Californians who were blind from childhood, are the best-known individuals to use these sonar techniques (Kremer, 2012; Kolarik et al., 2017).

In imaging studies, blind research participants show increased activity in visual cortical areas, and to some degree in the mid-temporal region, when they actively localize objects based on their acquired sonar methods. Such activation is strong in individuals who became blind early in childhood, weaker in people who lost their eyesight later in life, and absent in normal control participants. Surprisingly, no differences are detected in the auditory cortex. These observations imply that the visual cortex, deprived of its natural input, can be trained to adopt a novel function, such as echolocation.³⁷ To succeed, self-initiated action is a precondition.

Olfaction

Smelling provides an elegant example of active sensing in mammals. In olfactory research, every investigator knows that whatever questions they study, the phase of respiration should be taken into account because respiratory movement modulates perception. The goal of sniffing is to optimally sample the odorant and route it to the olfactory receptors. Sniffing is a rhythmic pattern (4–10 sniffs per second) that generates turbulence in the nose so that odorants, which would only slowly diffuse during regular breathing, come in direct contact with the receptor. As a result of sniffing, the sampling becomes intermittent: effective during inhalation and reduced or absent during exhalation. But that is not all. The entire olfactory system gets informed ahead of time about the expected arrival of the odorant by a corollary discharge signal, initiated by exhalation (or more likely due to relaxation of the inspiration motor action). While the exact paths and mechanisms of this signaling are not well understood, ample physiological evidence shows that the entire system, from the olfactory receptor neurons to the olfactory (piriform) cortex, generates rhythmic patterns in phase with the sniff cycles. From the olfactory receptor neurons to the mitral cells (the main output of the olfactory bulb), all neurons show large-amplitude membrane voltage oscillations and synchronous spiking to the sniff cycle. Pyramidal neurons in the olfactory cortex, where the significance and meaning of the odor information are likely extracted, also show strong inhalation-coupled spiking output.

One may argue that the mechanical effect of active sniffing, inducing turbulence in the nose, is sufficient to enhance odorant detection and identification without any need to inform neurons in the olfactory system about the phase of respiration. To counter that argument, researchers injected odorants into the

37. De Volder (1999); Thaler et al. (2011).

bloodstream at low concentrations in human volunteers, thus removing the air turbulence effect. Participants still detected the odorant more reliably when they sniffed, demonstrating that, even in the absence of the odorant in the nose, the motor action still provides a gain.³⁸ This finding supports the claim that feedback from action to the sensory system enhances perception.

Vision as a Searchlight

Corollary discharge from saccadic eye movements serves as a signal to inform the visual system whether the world or the eyeball is moving. The same circuit is involved in visual search, which is another name for active sensing. Vision or seeing is not a passive camera-like function but an active scanning of the environment.³⁹ High-acuity vision in humans (i.e., what allows you to read text in this book) is possible only with the cone receptors of the foveal region, corresponding to the central five degrees of the field of vision. One degree of visual angle is roughly the width of your index finger held at arm's length. If we fix our eyes on some part of a picture to keep the high-acuity receptors in place, we can see that the surrounding region becomes blurry. The effect is like pointing a flashlight toward part of the environment in darkness. To see what is out there, we must move the flashlight back and forth. During the day as well, we are taking in our surroundings in snippets, but because of the frequent scanning eye movements, our visual system's memory creates the illusion that we see the entire visual scene with high acuity and at once.

Even when we fix our eyes on a spot, eye movements assist vision via frequent “microsaccades,” tiny jumps of the eye that move the scene across a few dozen photoreceptors. These movements ensure that our eyes do not fixate on a completely stationary picture. Microsaccades increase visual performance by enhancing the spiking activity of neurons in the visual cortex. When macaque

38. The authors (Bocca et al., 1965) interpreted their observations showing that the mechanical stimulation of the epithelium by sniffing lowered the threshold for detection of blood-borne molecules. A contemporary demonstration of bypassing the olfactory mucosa is by Dmitri Rinberg and colleagues (Smear et al., 2013). They optogenetically stimulated a single glomerulus, the functional unit of olfactory processing, and showed that this “implanted odorant” was sufficient for odor perception by simultaneously relaying identity, intensity, and timing signals.

39. I have often wondered where visual science would stand today if early visual researchers had been engineers or control theorists. Instead of taking the reductionist path of fixing heads and eyes, what if they had developed methods for detecting the position of the eyes in relationship to the environment in freely behaving animals and observed the mechanisms of seeing in action? It is notable that David Hubel focused his interest on the significance of eye movements in vision toward the end of his career (Martinez-Conde et al., 2004).

monkeys fixate on a target, and an optimally oriented line is centered over a cell's so-called receptive field⁴⁰ in the primary visual cortex, spiking activity increases after microsaccades. Importantly, such microsaccade-induced, transiently enhanced neuronal excitability is present in every visual area examined by different research groups and even in the lateral geniculate body, a thalamic structure that transmits the retinal information to the primary visual cortex. A great number of neurons are affected by eye movements, as might be expected because visual areas occupy a large part of the brain.⁴¹ In other words, movements of the eye improve the visual performance of the sensory system and assist with the maintenance of perception during visual fixation. Saccades thus can be considered “visual sniffs,” which provide a sensory gain (Chapter 11).

In terms of visual function, there is no real difference between regular and microsaccades. During eye fixation, the saccades are smaller (“micro”), while during “voluntary” visual search, they are larger. Both types of saccades are initiated in the superior colliculus in the midbrain. The widespread anatomical projections from this structure may explain why AI could effectively compensate for her loss of eye muscles by inducing saccade-like head movements (Figure 3.2). This observation emphasizes an important rule in neuronal organization: functions that are important for survival are widely and redundantly distributed in the brain—a robust safety mechanism.

Hearing

When our ears are plugged, we have trouble not only hearing others but also speaking or singing properly. Auditory feedback is very useful in speech.

40. Neurons respond to various constellation of inputs. The most robust response is believed to reflect the “relevant” input. Many visual neurons have preferred orientation selectivity, motion direction selectivity, color preference, etc., which we call the receptive field.

41. Many excellent original studies on eye saccades are compressed in these short paragraphs. Some of these studies are reviewed by Yarbus (1967) and Carpenter (1980) and Martinez-Conde et al. (2004). Martinez-Conde et al. emphasized the importance of eye movements in vision but considered them only to affect retinal processing: “microsaccades primarily modulate neural activity in early visual areas through retinal motion.” Corollary discharge within the brain as a mechanism for visual gain is not discussed, even though saccades can affect the entire visual system. See also Otero-Millan (2008). Hofmann et al. (2013) discusses the many forms of active sensing in bats (echolocation) and invertebrates (e.g., electric communication). Wachowiak (2011) and Morillon et al. (2015) are excellent reviews on the role of motion in hearing and olfaction, respectively. Visual saccades affect even hippocampal neurons (Meister and Buffalo, 2016).

Distinguishing between our own speech and a recording of it is a piece of cake because the corollary discharge mechanism provides an intimate relationship between sound production and hearing. Each uttered sound leads to the next action, and every sound is preceded by action preparation. Sound utterances adjust the sound-controlling muscles and help to achieve the desired outcome.

As in other sensory systems, the auditory regions of the cortex also receive direct projections from motor areas, providing a substrate for conveying corollary discharge signals. A copy of the motor command from the premotor cortical region is sent to the auditory cortical areas, where it strongly suppresses both spontaneous and tone-evoked synaptic activity during a wide variety of natural movements. This is achieved by a so-called *feed-forward inhibitory mechanism*, which entails the activation of inhibitory interneurons in the auditory cortex and the consequent transient inhibition of the spiking activity of pyramidal neurons. As a result, sound-evoked cortical responses are selectively suppressed during movement.⁴² In addition to cortical mechanisms, the muscles of the middle ear, which regulate the sound transmission from the eardrum to the cochlea, perform a similar preemptive action. Their activation by the movement command strongly attenuates the sound pressure before it can affect the cochlea. These corollary actions of the brain's output are responsible for making the sounds of the heavy timpani in Handel's *Messiah* pleasing, whereas the same sounds encountered unexpectedly would startle us.

Action-induced corollary signaling thus can transiently dampen the sensitivity of the auditory system to predictable sounds, while maintaining its responsiveness to unexpected stimuli. These same mechanisms, when derailed by disease, are implicated in tinnitus and auditory hallucinations.⁴³ Neuronal excitation sent from the motor areas to the auditory areas that does not occur during self-induced sounds are interpreted by the brain as real.

Body Sensation

The traditional separation of motor and sensory functions is even more prominent in the somatosensory system. This view is explicitly illustrated by the classical textbook figures of motor and sensory “homunculi” whose jobs are to perceive and act, respectively, with a one-way, sensory-to-motor

42. Paus et al. (1996); Houde and Jordan (1998); Zatorre et al. (2007); Eliades and Wang (2008); Nelson et al. (2013); Schneider et al. (2014).

43. Feinberg et al. (1978); Ford and Mathalon (2004); Langguth et al. (2005); Nelson et al. (2013).

communication between them.⁴⁴ But recent discoveries in rodents paint a different picture. Rodents have an elaborate whisker system adapted for tactile exploration, especially in the dark. Whiskers palpate objects just like fingers do. There are multiple loops between the input and output paths, which include short brainstem connections and longer thalamic, cortical, and cerebellar paths. Importantly, ample anatomical evidence shows that the traffic between sensory and motor areas is bidirectional even in the cortex. Signals of whisker motion reach the somatosensory area and can serve as corollary discharge. These signals enable neurons in this area to integrate the movements and touches of multiple whiskers over time, key components of object identification and navigation by active touch. In addition, the somatosensory area of whisker representation (called the *barrel cortex*) forms an equally direct and prominent motor control pathway. Activation of neurons in the “somatosensory” cortex is as effective in controlling whisker movements as stimulation of the anatomically designated motor cortex. Direct anatomical projections from the somatosensory cortex to the spinal cord are present not only in rodents but in monkeys as well, suggesting that similar mechanisms are at work in different species.⁴⁵

A classical demonstration for the critical role of action in perception is the tactile-vision sensory substitution experiment by Paul Bach-y-Rita and colleagues at the University of Wisconsin, which allowed blind people to “see” using an array of electrodes placed on the tongue and stimulating the sensory terminals with weak electrical pulses. Blind participants were able to partially gain some visual function conveyed to the tongue by the output of a video camera. A crucial component of the successful outcome in each case was the participants’ ability to control of the camera. Under passive tongue “viewing” conditions, only tickling the tongue was felt but vision-like sensation did not occur in the participants.⁴⁶

Of course, you may counter my arguments by simple introspection and say that you can sit completely immobile and yet perfectly process the sensory flow. A touch on your hand or a bug flying in your visual field can be detected with no muscular effort whatsoever. Therefore, such examples can

44. Neurosurgeon Wilder G. Penfield, working with neurologist Herbert Jasper at the Montreal Neurological Institute, electrically stimulated the brain of awake patients with electrical probes and observed their motor responses or obtained their verbal feedback. The stimulation technique allowed them to create crude sensory and motor maps of the cortex (cartoon representations of the body surface, or *homunculi*). These somewhat misleading anatomically discrete maps are still used today, practically unaltered.

45. Ferezou et al. (2007); Mátyás et al. (2010); Hatsopoulos and Suminski (2011); Huber et al. (2012); O'Connor et al. (2013); McElvain et al. (2017).

46. Bach-y-Rita et al. (1969).

be taken as evidence in favor of the sensory-to-motor sequence of perception. However, such perception occurs in an already “calibrated” brain. Without actively sensing stimuli, without investigating them at some point in life, stimuli cannot not acquire a meaning to the brain. Once meaning (i.e., significance of a stimulus or event for the animal) emerges, it can be placed in a memory store. We can say it becomes *internalized*, and the internalized pattern can function as a grounding mechanism for further perceptual interpretations. I will expand on these issues in Chapter 5.

BODY TEACHING BRAIN

Although the brain has no a priori clues about what its sensors are sensing or what its effectors are effecting (Chapter 1), the developing brain does not start from scratch; it benefits enormously from inherited and early programs. But one-size-fits-all blueprints are not adequate to do the job because bodies come in different shapes and forms. The genetic blueprint is simply a “protomap.” In the newborn rat, the spinothalamic tract and thalamocortical somatosensory pathways are already in place. This map thus has some correspondences with body parts. However, cross-talk among the body part representations is limited at this early stage because the corticocortical connections are just beginning to grow after birth. How do the growing axons “know” which neurons to innervate and which ones to avoid? The proximity relationships help but are not sufficient.

At the output end, the ventral horn motoneurons in the spinal cord, already wired to the muscles, begin to generate irregular, uncoordinated movements. These seemingly aimless movements in newborn rodents are the same as fetal movements or “baby kicks” observed in later stages of pregnancy in humans. Every expectant mother and physician knows that such kicks are important aspects of the normal development of the fetus. However, the biological utility of the kicks and their service to the brain have been clarified only recently. As you will see later, each kick helps the brain to learn about the physics of the body it controls.

Stretch sensors in muscles and tendons report the contractile state of the muscle to the spinal cord and eventually to the somatosensory cortex. In addition, twitches of skeletal muscles increase the probability that the skin over the muscle will touch another pup in the nest or touch the wall of the womb in case of human fetuses. Due to the physical constraints of the bones and joints, only a limited fraction of muscle movement combinations ever occurs, out of the potentially very large number of possibilities that could result from unrestrained combinations of muscle activity. All these movement combinations

are meaningful teaching patterns to the somatosensory thalamus and cortex because these are the combinations that will be used later in life. Random and independent induction of cortical activity by the 600 or so skeletal muscles that move the mammalian body would be of very little biological relevance. Furthermore, sending organized inputs to the brain cannot be achieved through merely a genetic blueprint because the metric relations among the body parts change rapidly as the body grows.

How can such dumb “training” from muscle twitch combinations contribute to the formation of the body map? In the newborn rat up, every twitch and limb jerk induces a “spindle-shaped” oscillatory pattern in the somatosensory cortex lasting for a few hundred milliseconds. In the adult animal, neurophysiologists would call such a pattern a thalamocortical “sleep spindle” because such patterns occur in adults only during non-REM sleep. Both in the pup and in a prematurely born human baby, these are the first organized cortical patterns.⁴⁷ When long-range corticocortical connections form after birth in the rat, the spindle oscillation can serve to bind together neuronal groups that are coactivated in the sensory cortical areas as a result of the simultaneous movement in neighboring agonistic muscles. Likewise, muscles with an antagonistic movement relationship in the body (e.g., the biceps and triceps of the upper arm) will induce consistent activity–silence relationships in their sensory cortex and create an inhibitory relationship between the respective neuronal groups. After the long-range corticocortical connections are in place a week or so after birth and the body map is formed, the spindles become confined to non-REM sleep. Then the pups can use their body map to respond to local touch and use the proprioceptive information from the muscles and muscle tendons for effortless ambulation.⁴⁸ Thus, the initially meaningless, action-induced feedback from sensors transduces the spatial layout of the body into temporal spiking relationships among neurons in the brain. This developmental process is how the brain acquires knowledge of the body it controls or, more appropriately, cooperates with. Thus, a dumb teacher (i.e., the stochastically occurring movement patterns) can increase the brain’s smartness about its owner’s body landscape. Once the body scheme is built, the relationship between spindles and muscle jerks disappears. The spindles become “internalized” (Chapter 5) and continue to occur during sleep as a self-organized pattern. Yet the relationship

47. Similar to early spindles, spontaneous retinal waves trigger activity in every part of the visual system during early developmental stages (Katz and Shatz, 1996).

48. Khazipov et al. (2004); Buzsáki (2006). The early spindles in premature human babies are called *delta-brush oscillations*, and the alternating pattern of silence and activity is known as *trace alternans*. As in rat pups, muscle twitches induce delta brushes at 29 to 31 weeks of postconceptional age in preterm neonates and in utero (Milh et al., 2007).

between movement and spindles does not completely vanish. The reader may have experienced occasional dramatic whole-body jerks just before sleep. Such movements trigger a sleep spindle, as in premature babies.

ACTION AFFECTS BODY MAPS

From my perspective, the term “somatosensory” cortex is a misnomer of the outside-in framework as it not only senses but also simulates the body.⁴⁹ This is most clearly demonstrated by the continued representation of missing body parts. The ownership of phantom limbs and the painful feelings that arises from “them” after their loss or amputation is a well-studied clinical problem. If the sensory cortex were truly sensory and fully dependent on external inputs, phantom limbs would not be constructed mentally.⁵⁰ Another striking example of body simulation by the somatosensory system is false body part illusions. In the laboratory, a research participant is seated with her left arm resting on a small table, hidden from her view by a screen, and a life-sized rubber model of a hand and arm is placed on the table directly in front of the participant. The participant is asked to fix her eyes on the artificial hand while the experimenter visibly touches the rubber hand with a paintbrush and, at the same time and unknown to the participant, also touches her hidden hand with another paintbrush. If the real and rubber arms are touched multiple times in synchrony, the average participant voluntarily accepts the rubber hand as her own. This simple procedure is sufficient to produce a feeling of ownership of a foreign body and incorporate it into the body scheme.⁵¹ In another version of the experiment, the experimenter moves the participant’s left index finger so that it touches the right rubber hand on the knuckle of the index finger, and, at the same time, the experimenter touches the participant’s right index finger on the knuckle. If the

49. Michael Brecht expanded on these ideas and formulated a body model theory of the somatosensory cortex (Brecht, 2017). He suggests that layer 4 of somatosensory cortex mirrors the entire body and not just sensory afferents. This body model is continuously updated by layer 6 to layer 4 inputs and compares to an avatar rather than a mere sensory map.

50. Ramachandran et al. (1995). Similarly, blind people can imagine and dream about visual scenes. Conversely, patients with bilateral damage to the primary visual cortex can still respond to some visual stimuli and navigate around visual objects, even though they report seeing nothing and having no memory of the objects. This phenomenon is known as *blindsight* (Cowey, 2010; see it yourself here: <https://www.youtube.com/watch?v=GwGmWqX0MnM>). Monkeys with primary visual area lesions behave similarly; Cowey and Stoerig, 1995). These observations illustrate that “sensory” cortices do not simply serve to veridically “represent” the external world.

51. See a demonstration here: <https://www.youtube.com/watch?v=sxwn1w7MJvk>.

touches on the two hands are synchronized, the participant soon feels that she is touching her own hand. The illusory touch, in turn, will activate the ventral premotor cortex, intraparietal cortex, and cerebellum, indicating that the illusion reflects the detection of congruent multisensory signals from one's own body, as revealed in imaging experiments.⁵²

These illusions derive from the lack of grounding by action. If the participant moves her hand and arm while the rubber hand is being touched, then no illusion arises. Furthermore, the participant loses the sense of ownership of the arm instantaneously when the experimenter moves it. An obvious direct test of the action-perception framework would be to deprive the brain entirely of its motor and autonomic outputs and examine whether sensation could be evoked in a brain that had never experienced touch in conjunction with action. Short of such an action-deprivation experiment, we could ask how the brain would handle an unusual body. For example, what if two brains shared one body? Nature has provided several such examples.

Two Brains Control One Body

Abby and Brittany are conjoined twins who live in Minnesota. They were charming children, and today they are self-confident adults. The twins have separate heads and spinal cords, a joint chest with two hearts, two hands, and two legs. Each brain and spinal cord innervates one hand and one leg. The sense of touch and limb control are restricted to the half of the body that each brain innervates. Yet Abby and Brittany can perform many movement patterns that require bilateral hand and leg coordination, such as walking, running, swimming, ball games, or driving a car. They have different tastes and even different career goals.⁵³ Because their brains have separate somatosensory innervation of parts of their shared body, many observers have wondered how they effectively coordinate their sensations to perform the synchronous or alternating movements needed for many simple or more complex activities. However, if action teaches perception, as I suggested in this chapter, then we can explain

52. Botvinick (2004); Ehrsson et al. (2005).

53. There are numerous YouTube sites on these twins; visit <https://www.yahoo.com/tv/conjoined-twins--abby---brittany--get-their-own-reality-show--video-.html>. Conjoined twins, when they are respectfully approached, could be a rich source of unique information on brain-body questions. Among these is the important question of whether a feeling of anxiety is a body-induced effect on the brain. For example, researchers could show anxiogenic pictures to one twin and can ask the other twin how she feels. Many other brain-body-brain questions could be explored in conjoined twins.

why movements of the “other body” are inevitably sensed by both brains. It is because the physical constraints of the skeletal system have enabled the incorporation of a body extension. The other body *becomes* part of the body scheme as a result of action-induced joint experience.

If action is so critical for perception, it should play a significant role in cognition as well. That this is indeed the case, I will discuss in Chapter 5. However, a prerequisite of that discussion is the understanding of the collective behavior of neurons, the topic of the next chapter.

SUMMARY

The short message of this long chapter is that perception is an action-based process, an exploration initiated by the brain. This message is fundamentally different from the representational view, fueled by the outside-in empiricist philosophy, which inevitably poses the question: What comes between perception and action? The homunculus with its decision-making power—and, I suspect, the problem of consciousness as well—are unavoidable logical consequences of the separation of perception from action.

I promote the alternative view: things and events in the world can acquire meaning only through brain-initiated actions. In this process, the brain does not represent the world in its numerous and largely irrelevant details but extracts those aspects that have become relevant to the organism by exploration. Thus the brain builds a simplified, customized model of the world by encoding the relationships of events to each other. These aspects of model building are uniquely different from brain to brain.

The critical physiological mechanism that grounds the sensory input to make it an experience is “corollary discharge”: a reference copy of a motor command sent to a comparator circuit from the action-initiating brain areas. This comparator mechanism allows the brain to examine the relationship between a true change in the sensory input and a change due to self-initiated movement of the sensors. The same corollary discharge mechanism also serves active sensing, the process by which sensory receptors can be most efficiently utilized to sense the environment.

